

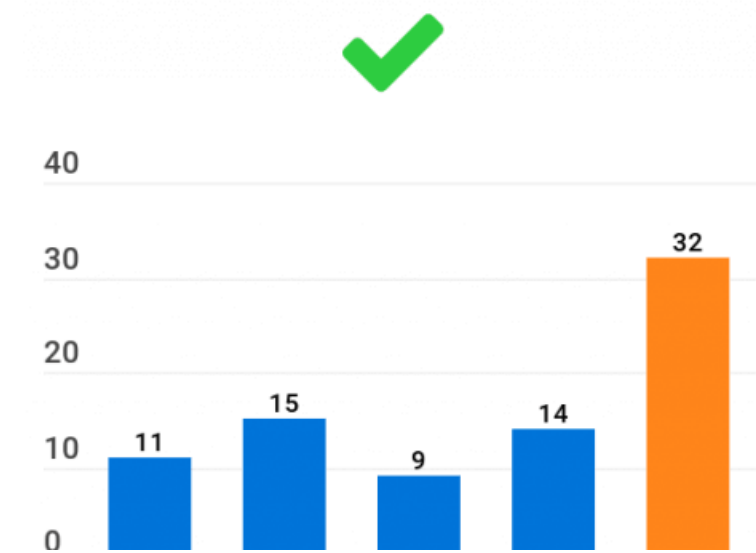
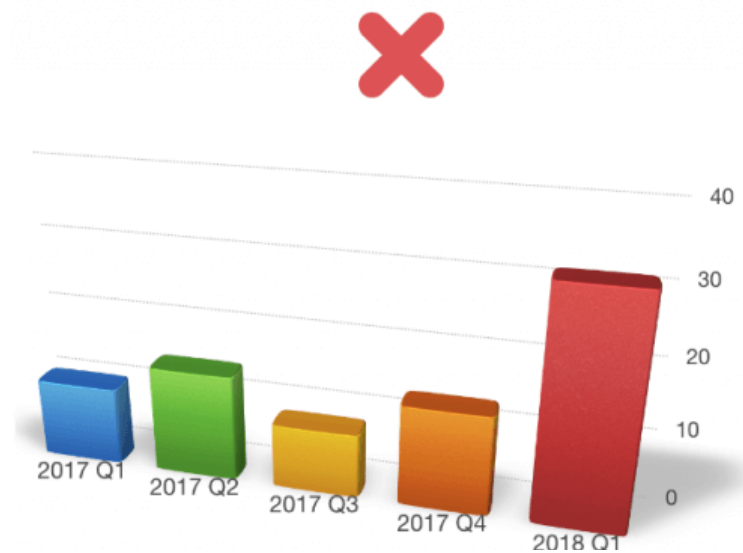
The power of visualization formatting

DATA VISUALIZATION IN DATABRICKS



Gang Wang
Senior Data Scientist

Why format your visualizations?



- The rainbow color theme makes the chart difficult to understand
- 3D imagery makes the data challenging to decipher

- Consistent bar type
- Simple color scheme
- Clear data labels

¹ Pictures: Infogram

Clear and descriptive axes

Chart with axis labels

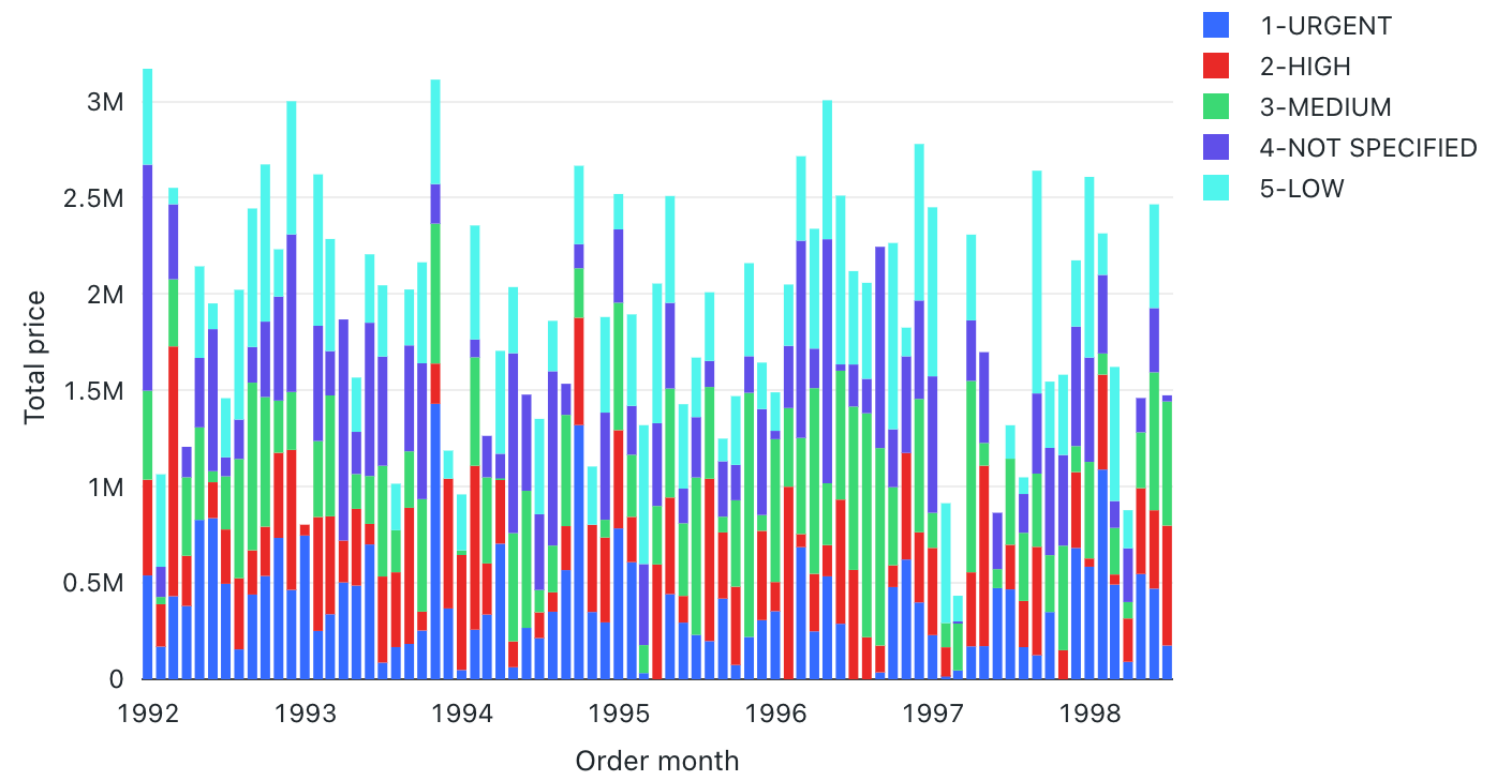
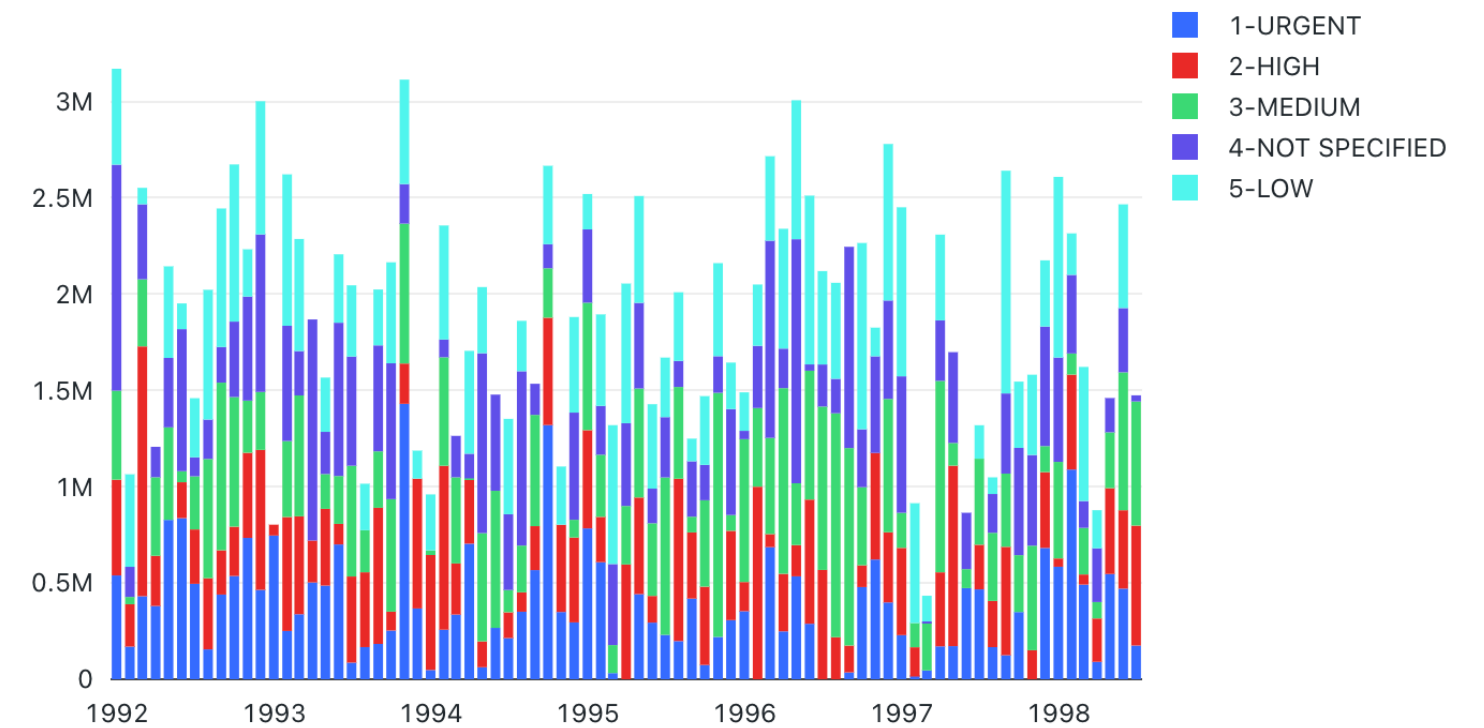


Chart without axis labels

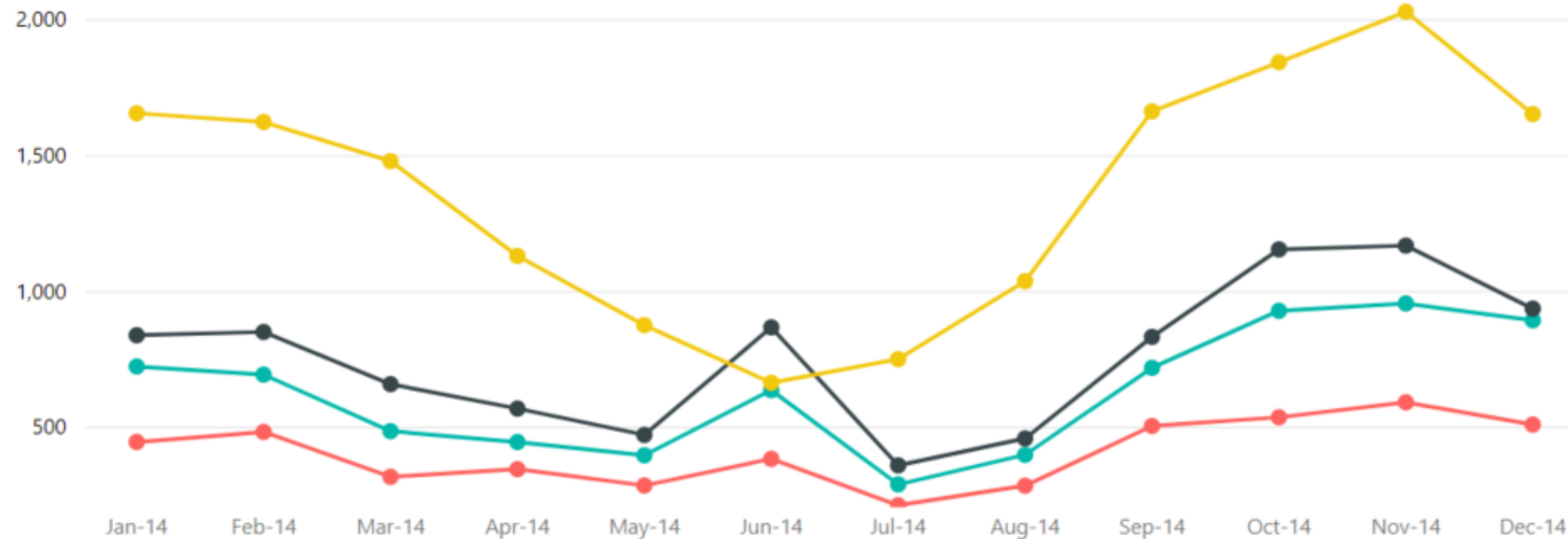


The role of color

- Distinguishes data series or categories
- Enhances readability and comparison

Total Units by Month and Manufacturer

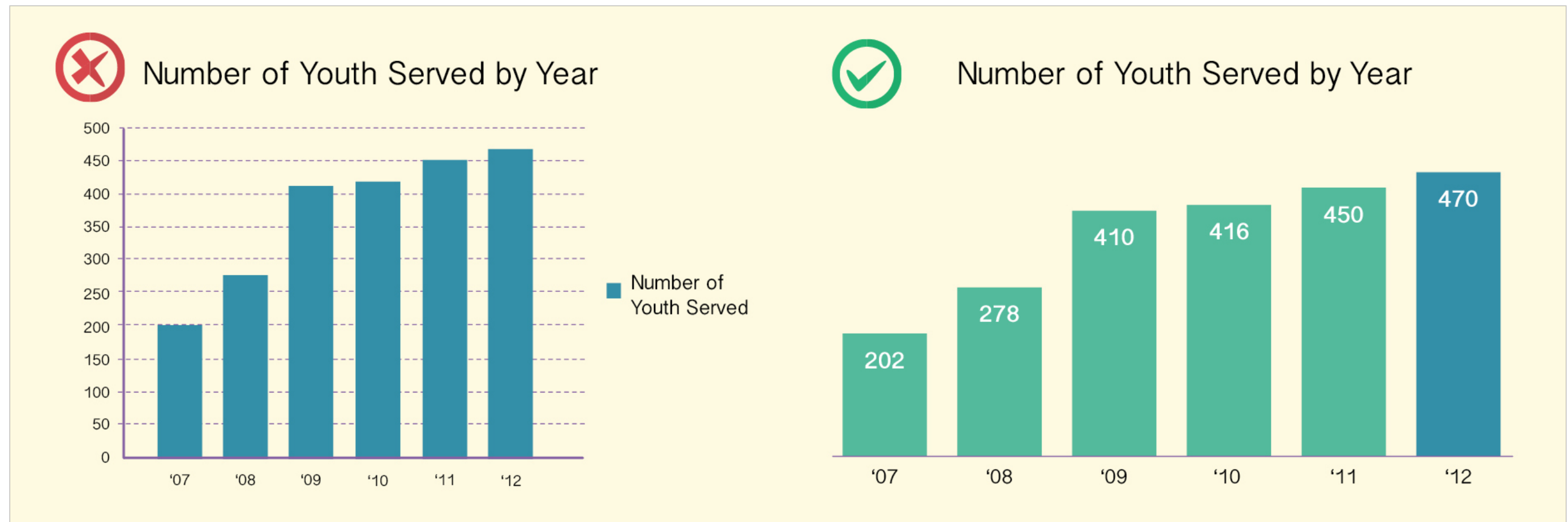
Manufacturer Aliqui Natura Pirum VanArsdel



¹ Pictures: Microsoft

Importance of data labels

- Provides precise information at a glance
- Helps in quick data interpretation



¹ Pictures: Visme

Summary of effective formatting

- Functionality over aesthetics
- Clarity
- Highlighting insights
- Audience engagement



¹ Pictures: Rock Content

Let's practice!

DATA VISUALIZATION IN DATABRICKS

Formatting visualizations in Databricks

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Senior Data Scientist

Let's practice!

DATA VISUALIZATION IN DATABRICKS

Data storytelling

DATA VISUALIZATION IN DATABRICKS



Gang Wang
Senior Data Scientist

What is data storytelling?

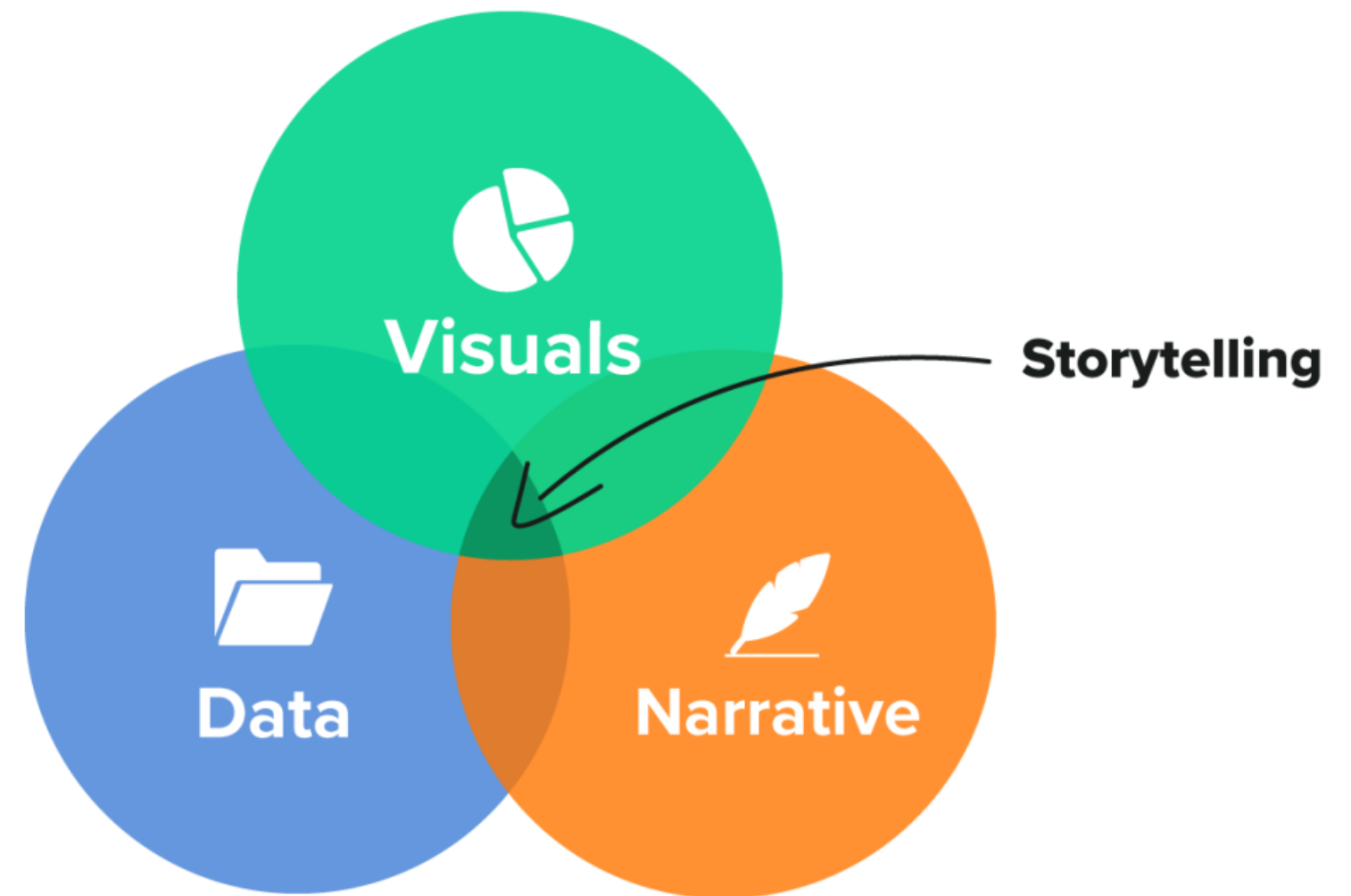
- Crafting narratives from data
- Combining insights with visualizations

Purpose:

- Make complex data accessible
- Engage and inform your audience

Why it matters:

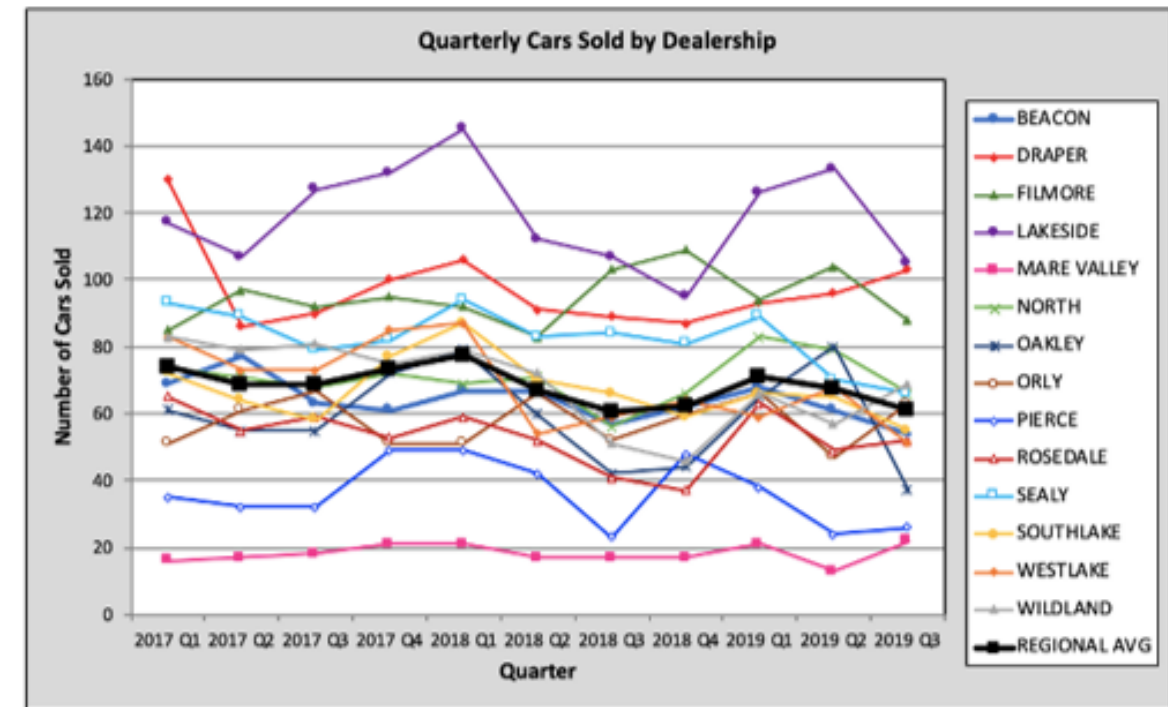
- Transforms raw data into insights
- Enhances decision-making



¹ Pictures: AgencyAnalytics

Before: showing data

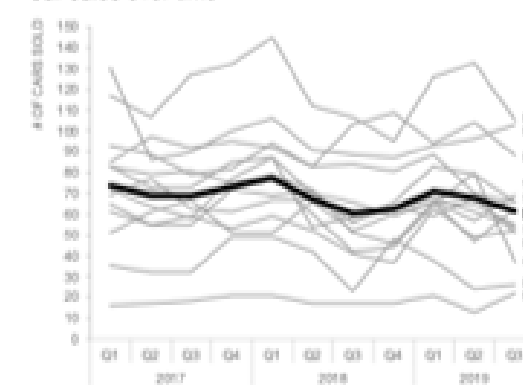
- Cluttered line chart, difficult to interpret
- Raw data without context



After: storytelling

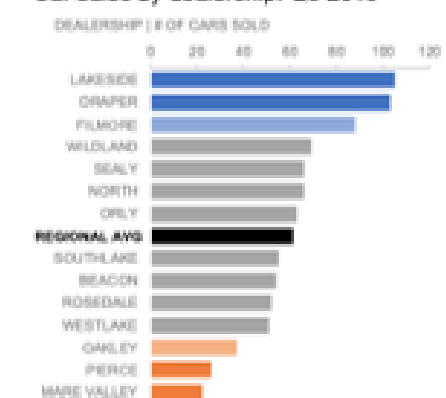
- Clear narrative highlighting key insights
- Simple bar chart emphasizing important data points

Car sales over time



Data source: Sales Database, includes cars sold onsite at regional dealerships through 9/30/19.

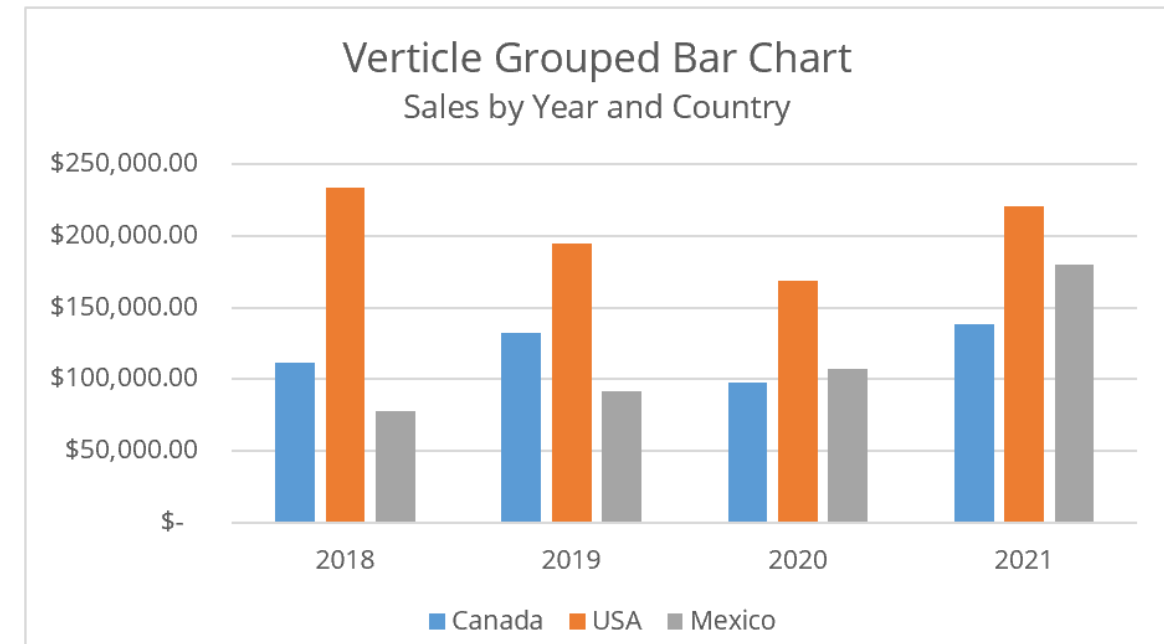
Car sales by dealership: Q3 2019



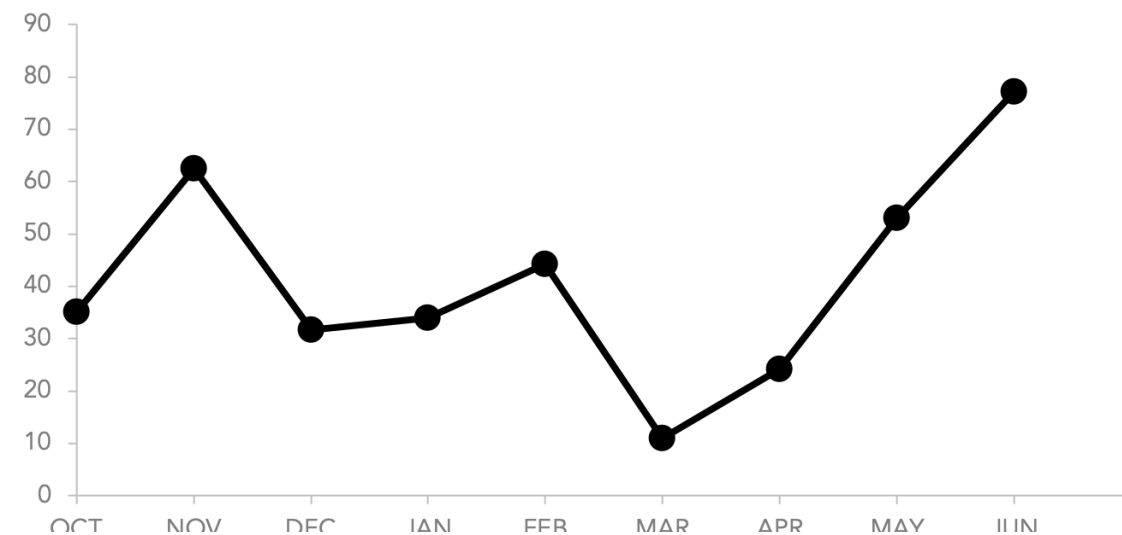
storytelling with data
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The role of data visualizations in storytelling

- Bar charts: compare quantities across categories
- Line charts: show trends over time



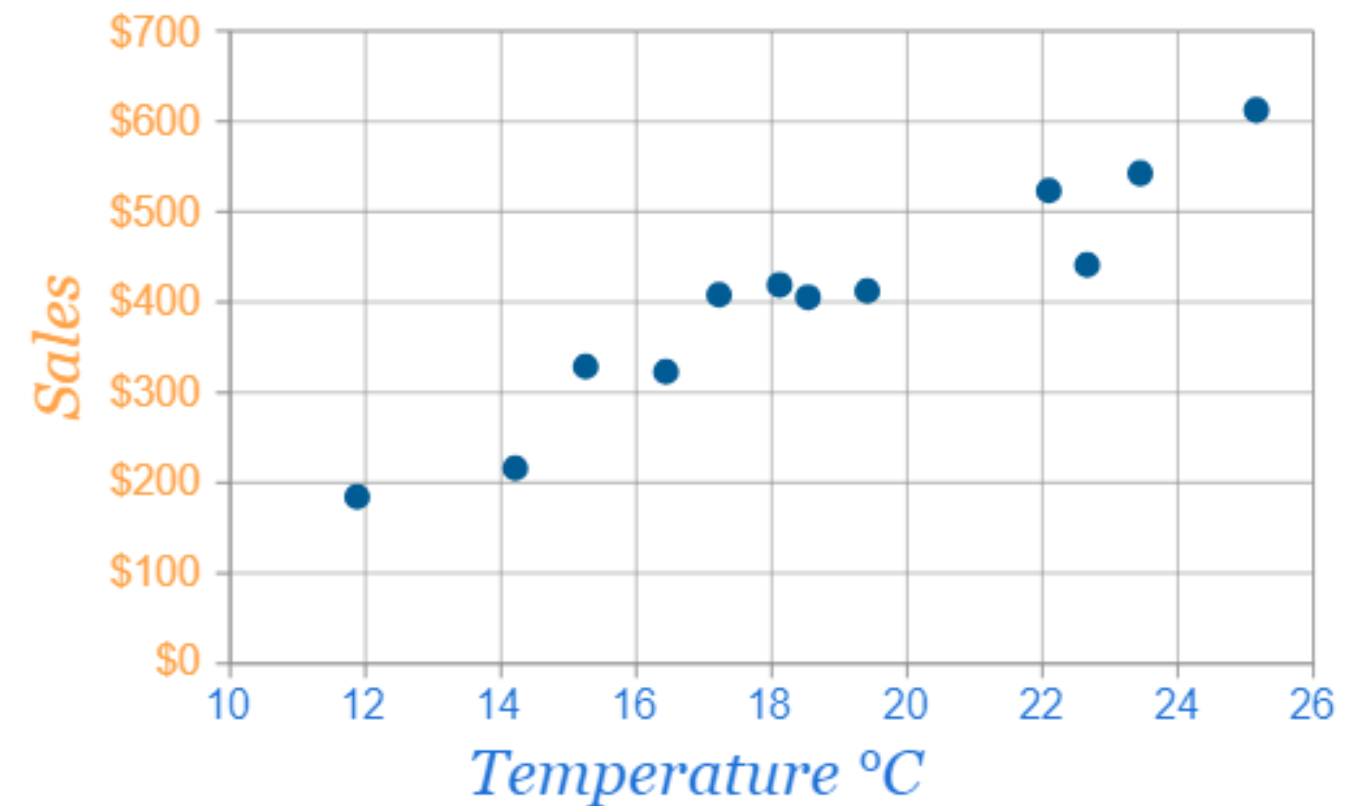
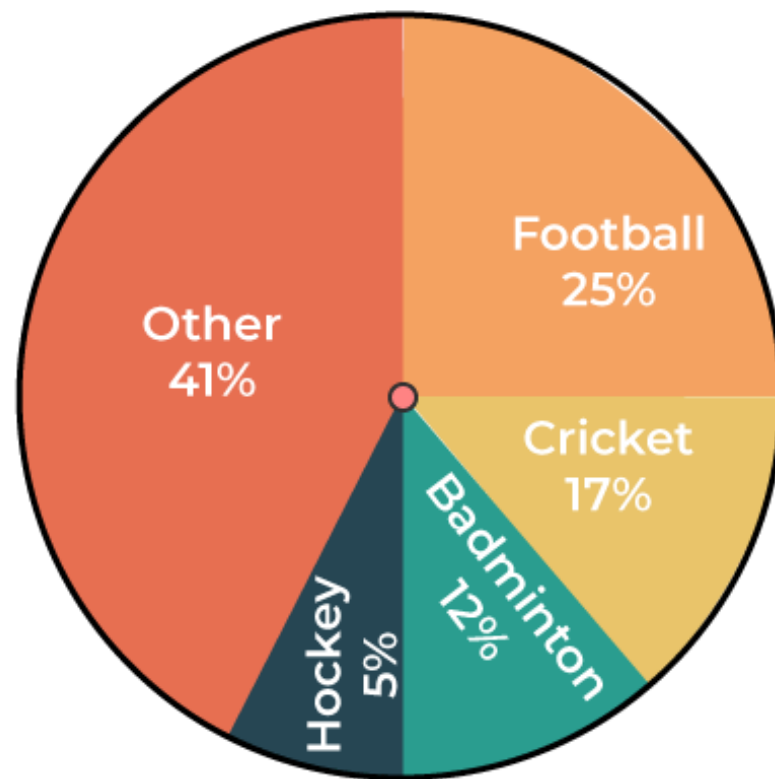
Produce sales
IN THOUSANDS (USD)



¹ Pictures: Corporate Finance Institute, Storytelling with data

- Pie charts: display proportions and part-to-whole relationships
- Scatter plots: show relationships and correlations

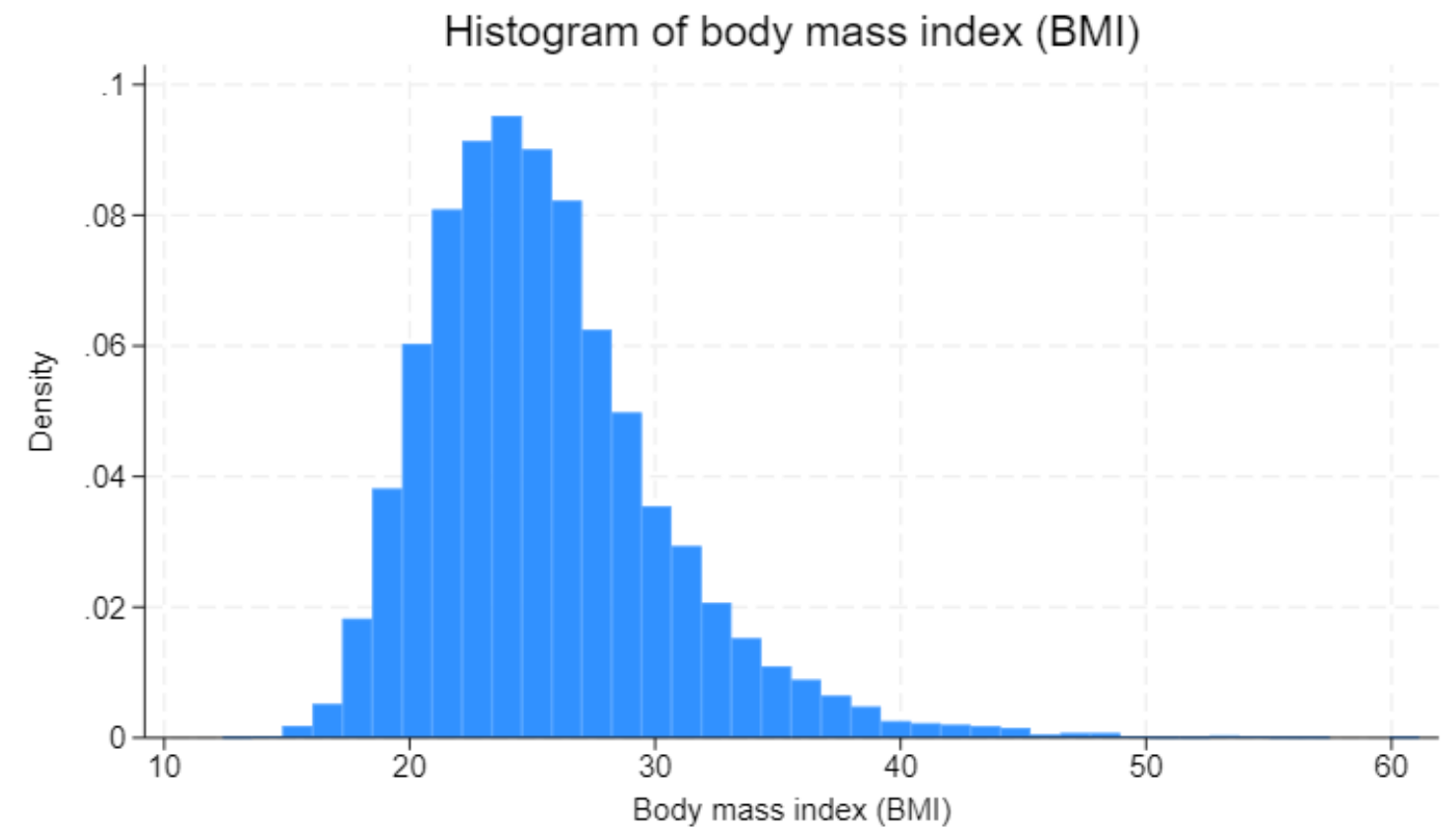
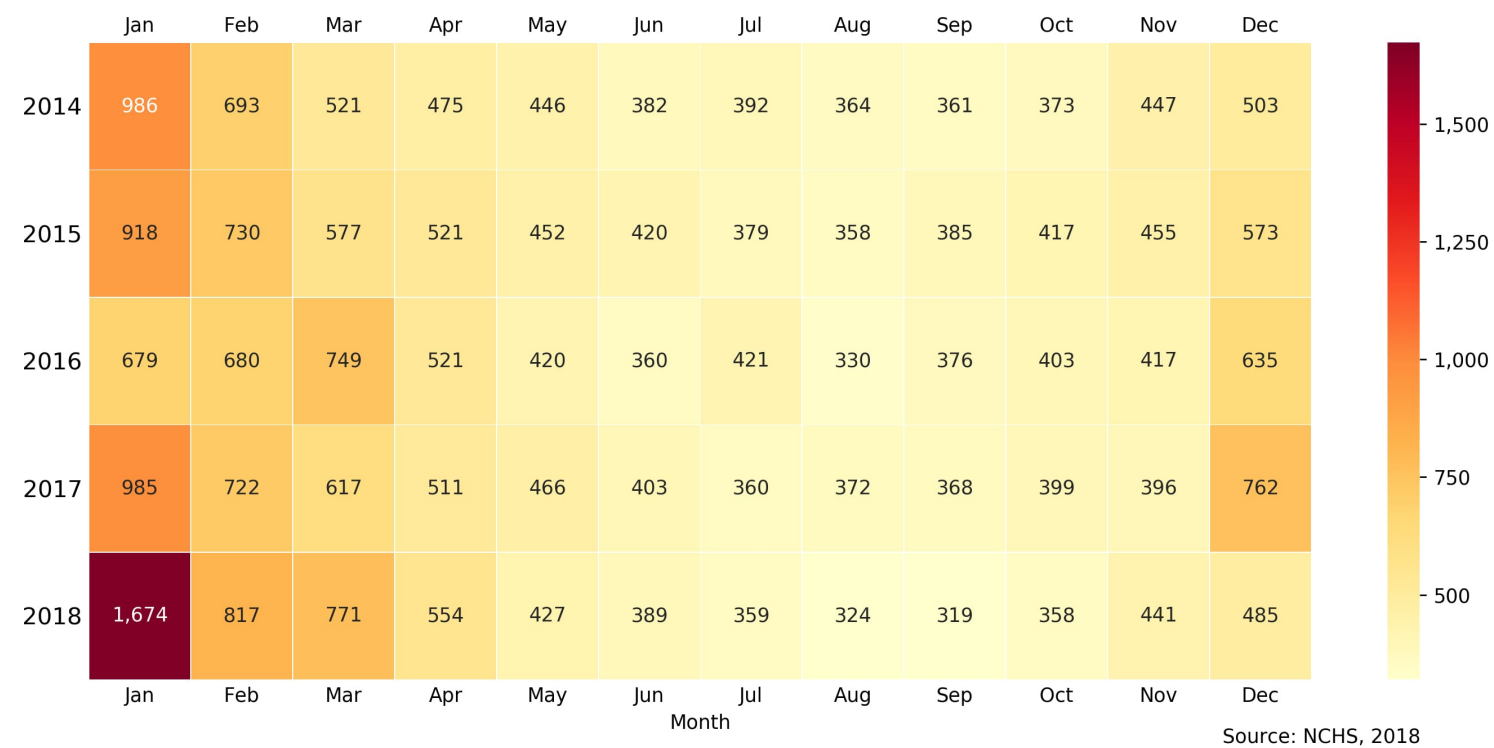
Number of Students



¹ Pictures: GeeksforGeeks, Math is Fun

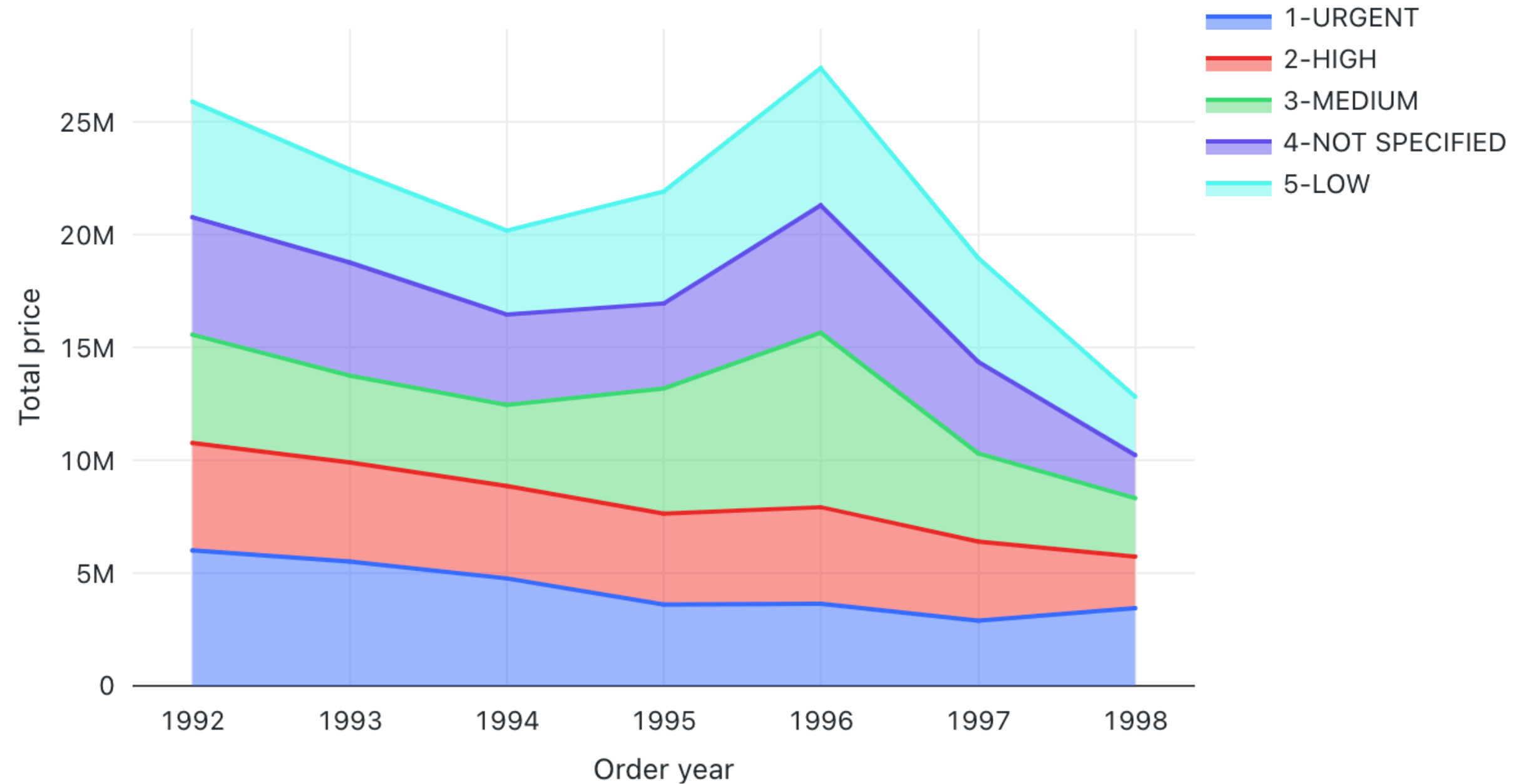
- Heat maps: reveal patterns and intensity of data
- Histograms: show the distribution of numerical data

Monthly deaths from influenza and pneumonia in California 2014-2018



¹ Pictures: Sage Research Methods, Stata

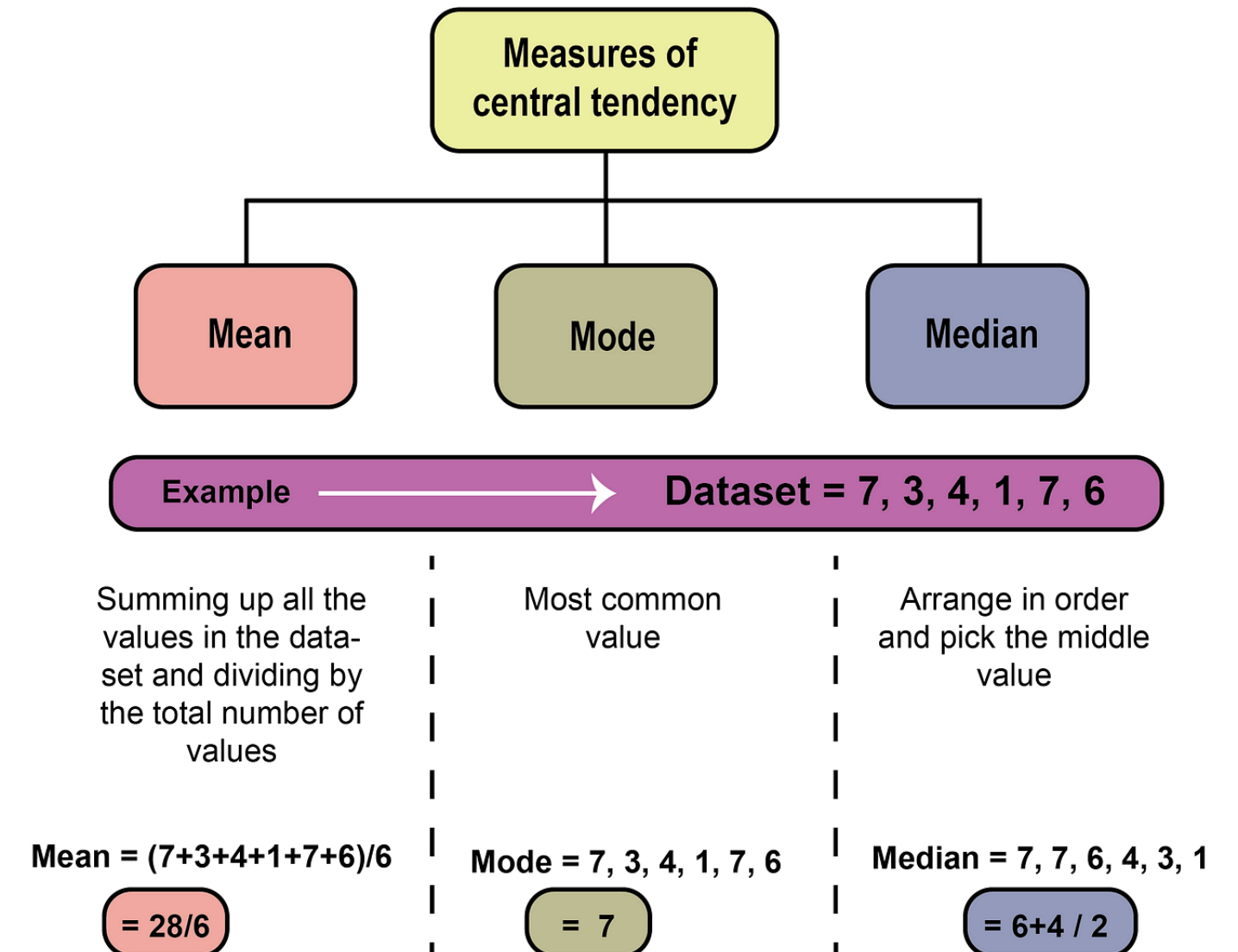
- Area charts: highlight changes over time with stacked data



Understanding statistical distributions

Why they matter

- Reveal patterns in data
- Highlight central tendencies: mean, median, mode
- Show spread: range, variance, standard deviation
- Identify shape: skewness, kurtosis



¹ Pictures: Medium

Creating Customizable Tables

Purpose of Customizable Tables:

- Tailor presentation to specific needs
- Highlight key insights clearly

Key features:

- Formatting options
- Column and row adjustments
- Sorting and filtering
- Data labels



¹ Pictures: UX Collective

Let's practice!

DATA VISUALIZATION IN DATABRICKS

Formatting Histograms

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Let's practice!

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